FRANKLIN YESUTOR FERGUSON

UAV Survey & Mine Technology · Digital Operations · Aerospace Engineer

CONTACT

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franklin's Portfolio

Mastiff ST GM-046-1558
Madina, Accra, Ghana

EDUCATION

BSc Aerospace Engineering
First Class Honours
KNUST, Kumasi, Ghana
Aug 2018 – Sep 2022
Provost Award ×3 for academic excellence

Google Cybersecurity Cert.
Coursera · Jan 2025

CORE COMPETENCIES

SURVEY & MAPPING

- RTK/PPK GNSS control & GCP design
- Orthomosaics, DTM/DSM & contours
- Stockpile, cut/fill & as-mined volumes
- Topographic & cadastral mapping
- LiDAR, multispectral & thermal data

FLIGHT OPERATIONS

- Hybrid VTOL fixed-wing & multirotor
- Tethered aerostat (RT SkyStar 180)
- BVLOS corridor & ground-spotter ops
- Flight-log analysis & failsafe set-up
- UAV maintenance, repair & build

DIGITAL & DATA

- Power BI dashboards & KPIs
- Power Apps, Power Automate, Planner
- SQL, Python, Node.js / Puppeteer
- GIS analysis in QGIS

RESEARCH PROFILE

First-Class Aerospace Engineer (KNUST) with extensive experience in Unmanned Aircraft Systems (UAS), aerial surveillance, security operations, geospatial intelligence, data analytics, and software development. Combines expertise in drone and aerostat operations, multisensor data acquisition, GIS, Power BI, and intelligence platform development to deliver actionable insights, strengthen operational security and intelligent geo-solutions. Has delivered technical solutions and executed projects for Newmont Ghana, Ghana Eximbank, UNDP, and the Ga East Municipal Assembly. Currently responsible for UAS operations, aerial surveillance, and security intelligence at Newmont Ghana's Ahafo Mine, leveraging aerial intelligence, data analytics, and digital platforms to enhance situational awareness and operational decision-making.

4+ years	~3,412 ha	Newmont	Tailored Solutions
Experience	Surveyed for Gexim 140 farms · 12 regions	Current Client	UNDP Mauritania

PROFESSIONAL EXPERIENCE

Security Intelligence & Systems Specialist — UAV & Aerostat Technician

Jan 2026 - Present
Current Role

Isen Ghana (Contracted to Newmont Ghana)
Newmont Ahafo Mines, Ghana

SURVEY AND MAPPING

- Run aerial and GNSS surveys end to end — RTK/PPK control, GCP layout, flight planning and processing into orthomosaics, DTMs, contours and volumes in Metashape, Pix4D and QGIS for mine planning, environment and security.
- Mapped three active pit areas (2026) and processed the data myself to expose illegal-mining access routes for site Security and the Mobile Response Unit (MRU).

SURVEILLANCE & SECURITY INTELLIGENCE

- Command live aerostat missions — deployment, EO/IR payload operation, real-time monitoring and safe recovery — to detect, classify and report illegal mining, briefing Newmont's command, Security and the MRU.
- Fly BVLOS corridor surveillance between Ahafo South and Ahafo North, backed by a ground-spotter training programme I wrote for the corridor.

SURVEILLANCE & SECURITY INTELLIGENCE

- Built a live Power BI security intelligence dashboard that brings detections, threat classifications, patrol summaries, incident trends and KPIs into one command view.
- Built a Power App for flight hours and logs, maintenance, downtime, helium top-ups and system updates, with automatic sync and dashboard refresh.
- Replaced paper inspection checklists with a web application: a backend generates PDF reports, sends them to stakeholders and archives every record automatically.

CORE COMPETENCIES

0PERATIONS & LEADERSHIP

- SOPs, risk assessments & incident reports
- Training design & delivery
- Team leadership & command briefings

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TECHNICAL TOOLKIT

FLIGHT CONTROL

ArduPilot (ArduPlane / ArduCopter), PX4, Mission Planner, QGroundControl, MAVLink, DJI

PLATFORMS FLOWN / BUILT

YANGDA Sky Whale VTOL, WingtraONE, DJI Drones, TT Aviation 30L, 50L, RT SkyStar 180 aerostat, custom multirotors

SURVEY & GIS

Emlid Reach RS2 GNSS, Agisoft Metashape, Pix4D, QGIS

DATA & SOFTWARE

Power BI, Power Apps, Power Automate, Excel, SQL, Python, Node.js, Puppeteer, Git

ENGINEERING DESIGN

SolidWorks, Fusion 360, Siemens NX, MATLAB, Arduino, ROS, 3D printing,

SECURITY & IT

Network-security fundamentals (Google Cybersecurity Certificate)

ISEN GHANA (Continued)

TRAINING & CAPABILITY

- Authored a ten-module UAV drone operator revision programme with a full reference manual.
- Designed and delivered a mapping and survey course for the aerostat team (Aug 2026)

Team Lead, Research & Development / UAV Engineer

SKT Live Limited · East Legon, Accra — Jan 2023 - Jan 2025

- **Led end-to-end UAV engineering and deployment**, configuring, testing, and operationalising multi-rotor and fixed-wing platforms for precision aerial applications, large-area surveying, and environmental monitoring.
- **Designed and engineered an autonomous seed-dispersal payload** for the TT Aviation 30L platform, integrating Arduino-controlled servo mechanisms, timed actuation logic, and mechanical components into a field-ready aerial delivery system.
- **Directed UAV subsystem design, 3D modelling, and prototyping** using SolidWorks and Fusion 360, applying iterative engineering and field testing to improve platform reliability and reduce operational failures.
- **Developed data analytics and performance dashboards** using SQL, Python, and Power BI, transforming large-scale aerial survey data into actionable insights and management-level reporting for the Tree Crops Development Authority (TCDA).
- **Led technical training and knowledge transfer** through Flying Labs, mentoring students in UAV technology, aerial robotics, autonomous systems, and STEM across multiple schools.

Co-founder & Chief Operating Officer

Jan 2023 - Jan 2025

Wingxtra Aerospace Limited · Accra

- Ran daily operations, financial planning and cash flow for a UAV engineering start-up.
- Led product development of drone systems for surveillance, agriculture and precision operations — from design and prototyping through testing and manufacturing to client delivery.
- Led a cross-functional engineering team and ran the test cycles that refined drone performance.

Aircraft Maintenance Intern

Jun 2019 - Jul 2021

School of Trade Training, Ghana Air Force Base · Accra

- Worked on aircraft systems, maintenance procedures and airworthiness checks on active Ghana Air Force aircraft.

SELECTED PROJECTS & FIELD OPERATIONS

Illegal-mining route mapping
Newmont Ahafo South · 2026

Apensu, Awonsu & Subika pit areas surveyed with UAV and RTK GNSS; processed in Metashape and QGIS.

RESULT Entry and exit route maps that Newmont Security and the MRU act on.

Security intelligence platform
Newmont Ahafo South · 2026

Power BI command dashboard, Power App flight & maintenance records, detection pipeline

RESULT One live view of detections, incidents and flight, maintenance status and shift's efficiency, refreshed automatically.

Nationwide Agricultural Intelligence Platform
Ghana Exim Bank

Led nationwide UAV surveys across 12 regions and 140 farms (~3,412 ha), capturing RGB, infrared, and LiDAR data for crop-health assessment, yield estimation, and agricultural investment-risk analysis. Collaborated with a cross-functional team to develop the end-to-end analytics pipeline, transforming aerial data into actionable agricultural intelligence.

RESULT Crop-health, yield and investment-risk evidence for the bank's agricultural lending.

Digital Revenue Intelligence System
Ga East Municipal Assembly

UAV Surveying · GIS · Data Platforms
Conducted property mapping and developed geospatial data infrastructure for a digital property-rate and revenue collection system, integrating digitised property records to strengthen revenue administration and compliance monitoring in collaboration with Arkadiah (Singapore).

RESULT Base map for a digital property-rate and revenue management system.

Dedekebe cadastral survey
SKT Live Limited · ~1,000 ha

Surveyed and mapped land parcels using the Wingtra ONE fixed-wing VTOL drone, delivering high-resolution orthomosaics, cadastral boundary mapping, and geospatial analysis

RESULT High-resolution orthomosaics and cadastral boundary maps.

Precision crop-spraying operations
SKT Live · Kundungu, Asamankese

Executed UAV crop spraying operations across large-scale farmland, delivering efficient agrochemical application and improving coverage consistency compared to manual methods.

RESULT More consistent coverage than manual application.

REFERENCES

Available upon request.